

Star life cycle

Stars are born in vast clouds of cold, dense interstellar gas and dust that evolve until, billions of years later, they run out of fuel and die.

The clouds that give birth to stars consist mainly of hydrogen gas. New stars are huge, spinning globes of hot, glowing gas—mainly hydrogen with some helium. Most of this material is packed into the stars' cores, setting off nuclear reactions—fueled by hydrogen—that form helium and release energy in the form of heat and light. When most of the hydrogen is used up, stars may fade away, expand, or collapse in on themselves.

Birth, life, and death of a star

Stars start out their life as clouds of gas and dust, called nebulae. After millions of years, these clouds begin to pull inward because of the gravity of the gas and dust. As it is squeezed, the cloud heats up to form a young star, known as a protostar. If this reaches 27 million degrees Fahrenheit, it is hot enough to start nuclear fusion—the reaction needed for a star to form. The energy produced prevents a star from collapsing under its own weight and makes it shine. What happens when the fuel runs out and the star dies depends on how much dust gathered in the first place.

The sun has existed for about 4.5 billion years, and has burned about half of its hydrogen fuel.

1 Interstellar cloud
Stars are born in huge clouds of dense, cold gas and dust. A supernova explosion or star collision can trigger star birth.

2 Fragments form
The cloud breaks up into fragments. Gravity pulls the most massive and dense of these into clumps.

3 Protostar forms
Gravity pulls more material into the protostar's core. Density, pressure, and temperature build up.

4 Spinning disk
The material being pulled in starts to spin round, blowing out jets of gas.

5 Main sequence star
The core becomes so hot and dense that nuclear reactions occur and the star shines.

6 Planets form
Debris spinning around the star may clump together to form planets, moons, comets, and asteroids.

7 Stable star
The glowing core produces an outward pressure that balances the inward pull of gravity.

Death of a small star

Stars with less than half the mass of the sun, called red dwarfs, fade away slowly. Once the hydrogen in the core is used up, the star begins to feed off hydrogen in its atmosphere, shrinking—over up to a trillion years—to become a black dwarf.

Black dwarf

When all fuel is used up and its light is extinguished, the star becomes a cinder the size of Earth.

Star continues to shrink and fade.

Light intensity fades out.

Star begins to shrink.

Death of a medium-sized star

When a star with the same mass as our sun has used up its hydrogen (after about 10 billion years) nuclear fusion spreads out from the core, making the star expand into a red giant. The core collapses until it is hot and dense enough to fuse helium. When this, too, runs out, the star becomes a white dwarf, its outer layers spreading into space as a cloud of debris.